Lecture 11

Part of a pair: reduced states and purification

Objectives. *The two spins of hydrogen in their ground state at zero field, ψ^-, are entangled, so the electron alone is in no pure state. This lecture finds the density matrix that describes one part of a pair. The electron of ψ^- turns out to be the unpolarised beam of Lecture 3, and no operation on the proton changes the electron's statistics unless its outcome is recorded and used. Conversely, every density matrix is the state of one part of a pure pair. The entropy of a part measures how entangled a pure pair is. A system at a temperature has a mixed state, the thermal state. The lecture ends with the one place where the course leaves ideal measurement: a probe read ideally can give a reading of the system that is not ideal.*

By the end of this lecture you will be able to:

- Find the reduced density matrix of one part of a pair, by the partial trace.
- Write a pure state of a pair in Schmidt's form, and find its entanglement entropy.
- Find the first system's state after a reading of the second system, with the outcome recorded and with it not recorded.
- Find the probabilities of a probe's readings and the system's state after each, from the coupling of system and probe.

11.1 The state of a part

Let a pair of systems, with the state spaces $\mathbb{C}^n$ and $\mathbb{C}^m$, be in a state with the density matrix ρ on $\mathbb{C}^n \otimes \mathbb{C}^m$. The pair may be in a pure state, $\rho = |\psi\rangle\langle\psi|$, or be an ensemble of pairs, as in Lecture 3. Theorem 3.4 applies to the pair as to any system. A reading of an observable $A = \sum_a a\, E_a$ of the first system alone is a reading of $A \otimes \mathbb{1}$ (Postulate 10.1), and it gives a with probability $\mathrm{Tr}\big(\rho\,(E_a \otimes \mathbb{1})\big)$. We look for a density matrix of the first system alone that gives these probabilities.

Theorem 11.1 (The state of a part). *Let ρ be a density matrix on $\mathbb{C}^n \otimes \mathbb{C}^m$.*

1. *There is exactly one $n \times n$ matrix ρ_1 with*
$$\mathrm{Tr}(\rho_1 A) = \mathrm{Tr}\big(\rho\,(A \otimes \mathbb{1})\big) \qquad \text{for every } n \times n \text{ matrix } A, \tag{11.1}$$
and it is a density matrix.
2. *Let K be a branch operator of the first system, a product of unitaries and projections on $\mathbb{C}^n$ as in (3.3). The branch $K \otimes \mathbb{1}$ of the pair has the probability $\mathrm{Tr}(K\rho_1 K^\dagger)$. If this is positive, the first system is afterwards described by $K\rho_1 K^\dagger/\mathrm{Tr}(K\rho_1 K^\dagger)$.*
3. *If $\rho = |\psi\rangle\langle\psi|$ and ψ has the coefficient matrix C, then $\rho_1 = CC^\dagger$.*

Proof.

1. *Uniqueness.* If $\operatorname{Tr}(XA) = 0$ for every A, take for A the matrix $|e_k\rangle\langle e_i|$. Then $\operatorname{Tr}(XA) = \langle e_i, Xe_k\rangle = X_{ik}$, so $X = 0$. Two matrices with the property (11.1) have a difference X of this kind.
   Existence. Put $(\rho_1)_{ik} = \sum_j \langle e_i \otimes e_j, \rho\,(e_k \otimes e_j)\rangle$. Then $\operatorname{Tr}(\rho_1 A) = \sum_{i,k}(\rho_1)_{ik}A_{ki}$, and
   $$\operatorname{Tr}\big(\rho\,(A \otimes \mathbb{1})\big) = \sum_{i,j} \langle e_i \otimes e_j, \rho\,(Ae_i \otimes e_j)\rangle = \sum_{i,j,k} A_{ki} \langle e_i \otimes e_j, \rho\,(e_k \otimes e_j)\rangle\,,$$
   which is the same sum.
   A density matrix. The adjoint $\rho_1^\dagger$ has the property (11.1) too, since $\operatorname{Tr}(\rho_1^\dagger A) = \overline{\operatorname{Tr}(\rho_1 A^\dagger)} = \overline{\operatorname{Tr}(\rho(A^\dagger \otimes \mathbb{1}))} = \operatorname{Tr}\big(\rho\,(A \otimes \mathbb{1})\big)$, so $\rho_1^\dagger = \rho_1$. For a unit vector v, the matrix $|v\rangle\langle v|\otimes\mathbb{1}$ is a projection, so $\langle v, \rho_1 v\rangle = \operatorname{Tr}\big(\rho\,(|v\rangle\langle v|\otimes\mathbb{1})\big)$ is a probability, and $\langle v, \rho_1 v\rangle \geq 0$. With $A = \mathbb{1}$, $\operatorname{Tr}\rho_1 = \operatorname{Tr}\rho = 1$. By the spectral theorem (*Linear Algebra*, Thm 10.7), $\rho_1 = \sum_k \lambda_k |u_k\rangle\langle u_k|$ with orthonormal u_k and eigenvalues $\lambda_k = \langle u_k, \rho_1 u_k\rangle \geq 0$ that add up to 1. Leaving out the terms with $\lambda_k = 0$, ρ_1 is the density matrix of the ensemble $\{(\lambda_k, u_k) : \lambda_k > 0\}$.
2. By Theorem 3.4, the branch has the probability $\operatorname{Tr}\big((K \otimes \mathbb{1})\rho\,(K^\dagger \otimes \mathbb{1})\big) = \operatorname{Tr}\big(\rho\,(K^\dagger K \otimes \mathbb{1})\big) = \operatorname{Tr}(\rho_1 K^\dagger K) = \operatorname{Tr}(K\rho_1 K^\dagger)$, by the cyclicity of the trace and (11.1). Write Prob for this probability. Afterwards the pair has the density matrix $\rho' = (K\otimes\mathbb{1})\rho\,(K^\dagger\otimes\mathbb{1})/\text{Prob}$, and for every A,
   $$\operatorname{Tr}\big(\rho'(A \otimes \mathbb{1})\big) = \frac{\operatorname{Tr}\big(\rho\,(K^\dagger AK \otimes \mathbb{1})\big)}{\text{Prob}} = \frac{\operatorname{Tr}(\rho_1 K^\dagger AK)}{\text{Prob}} = \frac{\operatorname{Tr}(K\rho_1 K^\dagger A)}{\text{Prob}}.$$
   By the uniqueness in the first part, the first system is described by $K\rho_1 K^\dagger/\text{Prob}$.
3. By Proposition 10.2, $(A \otimes \mathbb{1})\psi$ has the coefficient matrix AC, so $\langle\psi, (A \otimes \mathbb{1})\psi\rangle = \sum_{i,j} \overline{C_{ij}}(AC)_{ij} = \operatorname{Tr}(C^\dagger AC) = \operatorname{Tr}(CC^\dagger A)$.

∎

Definition 11.2 (Reduced density matrix)**.** The matrix ρ_1 of Theorem 11.1 is the *reduced density matrix* of the first system, written $\rho_1 = \operatorname{Tr}_2\rho$. The map $\rho \mapsto \operatorname{Tr}_2\rho$ is the *partial trace* over the second system. The reduced density matrix $\rho_2 = \operatorname{Tr}_1\rho$ of the second system is defined by $\operatorname{Tr}(\rho_2 B) = \operatorname{Tr}\big(\rho\,(\mathbb{1} \otimes B)\big)$.

So every experiment on the first system alone, however long, is predicted by ρ_1. For such experiments, the second system can be forgotten. For a pure state the same computation, with $(\mathbb{1} \otimes B)\psi$ of coefficient matrix CB^T, gives $\operatorname{Tr}(C^\dagger CB^\mathsf{T}) = \operatorname{Tr}(C^\mathsf{T}\overline{C}B)$, so

$$\rho_1 = CC^\dagger, \qquad \rho_2 = C^\mathsf{T}\overline{C}, \tag{11.2}$$

where $\overline{C}$ has the complex conjugate entries. The subscripts 1 and 2 name the part, as t names the time in ρ_t.

Linear Algebra, Ex. 17.20, previews this. Nothing below depends on it.

11.2 A pure pair with mixed parts

Proposition 11.3 (Entangled pairs have mixed parts)**.** *A unit vector ψ of the pair is a product if and only if ρ_1 is pure, and if and only if ρ_2 is pure.*

Proof. For every vector v, $\langle v, CC^\dagger v\rangle = \|C^\dagger v\|^2$. So $CC^\dagger v = 0$ exactly when $C^\dagger v = 0$, the two matrices have the same kernel, and $\operatorname{rank}\rho_1 = \operatorname{rank}C^\dagger = \operatorname{rank}C$. A density matrix is pure exactly when its rank is 1. Indeed, a density matrix of rank 1 is $\lambda\,|u\rangle\langle u|$ with a unit vector u, and its trace gives $\lambda = 1$. By Theorem 10.4, ψ is a product exactly when $\operatorname{rank}C = 1$. The same argument with $\rho_2 = C^{\mathsf T}\overline{C}$ and $\operatorname{rank}\overline{C} = \operatorname{rank}C$ gives the second statement. ∎

For the ground state ψ^- of hydrogen's two spins at zero field, $C = \frac{1}{\sqrt2}\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$, and

$$\rho_1 = \rho_2 = \tfrac12\mathbb{1}.$$

The electron alone is unpolarised. A reading of its spin along any axis gives ±1 with probability $\frac12$ each. Its density matrix is that of the beam of silver atoms straight from the oven in Lecture 3. By Theorem 11.1 and Theorem 3.4, no experiment on the electron alone, and no sequence of experiments, tells the two apart.

Yet the two were prepared in different ways. The oven beam is an ensemble. Each atom is in some pure state, and we do not know which. The electron of ψ^- belongs to a pair that is in a pure state, and the electron alone is in none. A mixture that arises as one part of a pure pair is called an *improper mixture*, after d'Espagnat. What this difference means, if anything, is a question of interpretation, taken up in *Quantum Mechanics III: Foundations*. The predictions are the same.

11.3 How entangled: Schmidt's form and the entropy

How mixed the part of a pure pair is measures how entangled the pair is.

Theorem 11.4 (Schmidt's form)**.** *Let ψ be a unit vector of $\mathbb{C}^n\otimes\mathbb{C}^m$, and let $\rho_1 = \sum_i p_i\,|u_i\rangle\langle u_i|$ with an orthonormal eigenbasis $u_1,\dots,u_n$ of ρ_1. Then there are orthonormal vectors v_i of $\mathbb{C}^m$, one for each i with $p_i > 0$, such that*

$$\psi = \sum_{i:\,p_i>0}\sqrt{p_i}\,u_i\otimes v_i, \qquad \rho_2 = \sum_{i:\,p_i>0} p_i\,|v_i\rangle\langle v_i|\,.$$

So ρ_1 and ρ_2 have the same nonzero eigenvalues, with the same multiplicities, and there are at most $\min(n,m)$ of them.

Proof. Expanded in the basis u_i of the first factor, $\psi = \sum_i u_i\otimes w_i$ for some vectors w_i of $\mathbb{C}^m$. For $A = |u_j\rangle\langle u_i|$ we have $(A\otimes\mathbb{1})\psi = u_j\otimes w_i$, so $\langle\psi,(A\otimes\mathbb{1})\psi\rangle = \langle w_j, w_i\rangle$, and the defining property of ρ_1 (Theorem 11.1) gives

$$\langle w_j, w_i\rangle = \operatorname{Tr}(\rho_1\,|u_j\rangle\langle u_i|) = \langle u_i, \rho_1 u_j\rangle = p_i\delta_{ij}.$$

So $w_i = 0$ when $p_i = 0$, and the vectors $v_i = w_i/\sqrt{p_i}$ with $p_i > 0$ are orthonormal. For every $m\times m$ matrix B, $\langle\psi,(\mathbb{1}\otimes B)\psi\rangle = \sum_{p_i>0} p_i\,\langle v_i, Bv_i\rangle$, and the definition of ρ_2 (Definition 11.2) gives the second formula. At most m vectors of $\mathbb{C}^m$ are orthonormal. ∎

Definition 11.5 (Entropy). The *entropy* of a density matrix ρ with eigenvalues $p_1, \dots, p_n$ is

$$S(\rho) = -\sum_{k:\, p_k>0} p_k \log p_k,$$

with the natural logarithm. It is often written $-\operatorname{Tr}(\rho \log \rho)$. The letter S always carries its argument here, since S alone is a spin.

If the nonzero eigenvalues of ρ are r numbers all equal to $1/r$, then $S(\rho) = \log r$. So $e^{S(\rho)}$ is an effective number, generally not an integer, of equally weighted orthogonal states. It depends only on the eigenvalues, not on how the mixture was prepared.

Lemma 11.6. *For $x > 0$, $\log x \le x - 1$, with equality only at $x = 1$.*

Proof. The function $x - 1 - \log x$ has the derivative $1 - 1/x$, negative for $x < 1$ and positive for $x > 1$, so its minimum is its value 0 at $x = 1$. ∎

Proposition 11.7 (Bounds on the entropy). *Let ρ be a density matrix on $\mathbb{C}^n$ of rank r. Then $0 \le S(\rho) \le \log r \le \log n$, and $S(\rho) = 0$ if and only if ρ is pure.*

Proof. Each term $-p \log p$ with $0 < p \le 1$ is at least 0, and it is 0 only at $p = 1$. So $S(\rho) = 0$ exactly when one eigenvalue is 1 and the others are 0, that is, when ρ is pure. Exactly r of the p_k are positive, and $\log r \le \log n$ since $r \le n$. By Lemma 11.6 with $x = 1/(r p_k)$,

$$S(\rho) - \log r = \sum_{p_k>0} p_k \log \frac{1}{r p_k} \le \sum_{p_k>0} p_k \Big(\frac{1}{r p_k} - 1\Big) = \frac{r}{r} - 1 = 0.$$

∎

For a pure pair, Theorem 11.4 gives $S(\rho_1) = S(\rho_2)$, the *entanglement entropy* of ψ. It is 0 exactly when the pair is a product (Proposition 11.3), and it is at most $\log \min(n, m)$. The singlet has $\log 2$, the largest value for two qubits. This measures the entanglement of pure pairs only.

For two qubits this measure can be read from the coefficient matrix. The eigenvalues p_1 and $p_2 = 1 - p_1$ of $\rho_1 = CC^\dagger$ satisfy $p_1 p_2 = \det \rho_1 = |\det C|^2$. Take $p_1 \le \frac{1}{2}$. Then $p_1 = \frac{1}{2}\big(1 - \sqrt{1 - 4\,|\det C|^2}\big)$, which grows with $|\det C|$. The entropy grows with p_1, since its derivative $\log\big((1 - p_1)/p_1\big)$ is positive for $p_1 < \frac{1}{2}$. So the entropy grows with $|\det C|$, from 0 at $\det C = 0$ to $\log 2$ at $|\det C| = \frac{1}{2}$. For the ground state of Example 10.8, $|\det C|$ falls as the field grows, so its entropy falls from $\log 2$ towards 0.

11.4 What the other system can and cannot change

Can something done to the proton change what a reading of the electron gives?

Proposition 11.8 (No signalling). *For every density matrix ρ of the pair, the reduced density matrix ρ_1 is unchanged by a unitary $\mathbb{1} \otimes V$, and by a reading of an observable $\mathbb{1} \otimes B$ of the second system whose reading is not recorded.*

Proof. For the unitary, $\operatorname{Tr}\big((\mathbb{1} \otimes V)\rho\,(\mathbb{1} \otimes V^\dagger)(A \otimes \mathbb{1})\big) = \operatorname{Tr}\big(\rho\,(A \otimes \mathbb{1})\big)$, since $\mathbb{1} \otimes V^\dagger$ commutes with $A \otimes \mathbb{1}$ and the trace is cyclic. For the reading, the observables $A \otimes \mathbb{1}$ and $\mathbb{1} \otimes B$ commute, and by the fourth condition of Theorem 5.2 a reading of $\mathbb{1} \otimes B$ that is not recorded does not change the statistics of $A \otimes \mathbb{1}$. These statistics, for every A, fix ρ_1. ∎

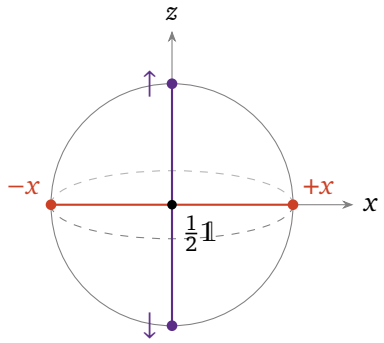


Figure 11.1: The electron of ψ^-. Unread, it is at the centre of the Bloch ball. A reading of the proton along z leaves it at one of the poles (each with probability $\frac{1}{2}$), and a reading along x at $\pm x$. Either pair of points, with weights $\frac{1}{2}$, has its centre of mass at the centre.

An interaction is not of this kind, and it can change ρ_1. In the evolution of Example 10.6, which starts from $|01\rangle$, (11.2) gives $\rho_1(t) = \operatorname{diag}(\cos^2(\nu t/2), \sin^2(\nu t/2))$. The electron's Bloch vector $(0, 0, \cos \nu t)$ passes through the centre of the ball at $\nu t = \pi/2$.

A reading of the second system that *is* recorded can change what we know about the first.

Example 11.9 (The proton prepares the electron)**.** Figure 11.1 shows the four states that occur. The pair is in ψ^-.

1. *The proton read along z.* With $E_\pm = \frac{1}{2}(\mathbb{1} \pm \sigma_z)$, the reading +1 has the probability $\|(\mathbb{1} \otimes E_+)\psi^-\|^2 = \frac{1}{2}$, and Lüders' rule leaves the pair in the ray of $(\mathbb{1} \otimes E_+)\psi^- = -\frac{1}{\sqrt{2}}\,|10\rangle$. The electron is down. The reading −1 leaves it up.
2. *The proton read along x.* In the same way the readings ± 1 have probability $\frac{1}{2}$ each and leave the electron in $|\mp x\rangle$, since $(\sigma_x \otimes \sigma_x)\psi^- = -\psi^-$ (§ 10.4).
3. *Recorded and pooled.* Each recorded reading prepares a pure state of the electron. Pooled, the readings along z give the ensemble $\{(\frac{1}{2}, |\uparrow\rangle), (\frac{1}{2}, |\downarrow\rangle)\}$ and those along x the ensemble $\{(\frac{1}{2}, |+x\rangle), (\frac{1}{2}, |-x\rangle)\}$. These are the two ensembles of Lecture 3, both with the density matrix $\frac{1}{2}\mathbb{1}$.

So a choice made on the proton, to read it along z or along x, decides which of two ensembles the electrons form. By Proposition 11.8 the choice cannot be seen in the electron's statistics. It shows when the electron's readings are sorted by the proton's readings, or in a reading of the pair together.

11.5 Every mixture is part of a pure pair

Lecture 3 met density matrices as ensembles. This lecture met them as parts of pure pairs. The two sources give the same objects.

Theorem 11.10 (Purification)**.** *Let ρ be a density matrix on $\mathbb{C}^n$. The vector ψ_ρ of $\mathbb{C}^n \otimes \mathbb{C}^n$ with the coefficient matrix $C = \sqrt{\rho}$, the positive square root of ρ, is a unit vector with*

$$\rho_1 = \rho, \qquad \rho_2 = \rho^{\mathsf{T}}.$$

If $\rho = \sum_k p_k\, |u_k\rangle\langle u_k|$ with orthonormal u_k, then $\psi_\rho = \sum_k \sqrt{p_k}\, u_k \otimes \overline{u_k}$.

Proof. By the spectral theorem, $\rho = \sum_k p_k |u_k\rangle\langle u_k|$ with $p_k \geq 0$, and $\sqrt{\rho} = \sum_k \sqrt{p_k}\, |u_k\rangle\langle u_k|$ (*Linear Algebra*, Thm 10.13). The matrix $|u\rangle\langle u|$ has the entries $u_i\overline{u_j}$, the coefficients of $u \otimes \overline{u}$, so $\psi_\rho = \sum_k \sqrt{p_k}\, u_k \otimes \overline{u_k}$. The $\overline{u_k}$ are orthonormal, so $\|\psi_\rho\|^2 = \sum_k p_k = 1$, and as in the proof of Theorem 11.4, $\langle \psi_\rho, (A \otimes \mathbb{1})\psi_\rho \rangle = \sum_k p_k \langle u_k, Au_k\rangle = \operatorname{Tr}(\rho A)$ and $\rho_2 = \sum_k p_k\, |\overline{u_k}\rangle\langle\overline{u_k}| = \overline{\rho} = \rho^{\mathsf{T}}$. ∎

A pure state of a pair whose first part has the density matrix ρ is a *purification* of ρ. For the unpolarised $\rho = \frac{1}{2}\mathbb{1}$, $\sqrt{\rho} = \frac{1}{\sqrt{2}}\mathbb{1}$, and the theorem gives $\Phi^+ = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. The singlet ψ^- purifies the same ρ, so a purification is not unique. Every density matrix of this course comes from an ensemble and from a part of a pure pair.

11.6 The thermal state

So far the mixed states of this lecture have been parts of pairs. Equilibrium gives another important mixed state. A system in equilibrium with surroundings at a temperature T has a density matrix that statistical physics supplies. We take it as given, since it does not follow from the partial trace alone. For a Hamiltonian $H = \sum_k \varepsilon_k E_k$ with distinct energies ε_k and $d_k = \operatorname{Tr} E_k$, and for $\beta > 0$, the *thermal state* is

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)} = \frac{1}{Z(\beta)} \sum_k e^{-\beta\varepsilon_k} E_k, \qquad Z(\beta) = \operatorname{Tr} e^{-\beta H} = \sum_k d_k e^{-\beta\varepsilon_k}, \tag{11.3}$$

with $\beta = 1/(k_B T)$ and k_B Boltzmann's constant (the boxes of Lecture 2 and Lecture 3 used $e^{-E/kT}$). The partition function $Z(\beta)$ always carries its argument, since Z alone is the observable Z of the three-level system of Lecture 4.

As a function of H, ρ_β commutes with H and is stationary (Theorem 4.4). It is a mixture of energy eigenstates, in which no Bohr frequency shows. A reading of the energy gives ε_k with probability $d_k e^{-\beta\varepsilon_k}/Z(\beta)$. Unless H is a multiple of $\mathbb{1}$, a temperature fixes the mean energy, not the energy. Exercise Sheet 11 shows that among all density matrices with the mean energy of ρ_β, ρ_β has the largest entropy.

11.7 A measurement that is not ideal

Every measurement so far has been ideal. A real detector reads a system through a probe coupled to it. Here the probe is read ideally, and the measurement it makes of the system is not ideal. We apply the rules for a pair to system and probe, and ask what the system alone sees, as in the rest of this lecture.

Example 11.11 (A probe). A system qubit is coupled to a probe qubit, prepared in $|0\rangle$, by the unitary

$$U_\theta = |0\rangle\langle 0| \otimes \mathbb{1} + |1\rangle\langle 1| \otimes e^{-\mathrm{i}\theta\sigma_y/2}, \qquad 0 \leq \theta \leq \pi,$$

and then only the probe is read, ideally, in the basis $|0\rangle, |1\rangle$.

1. *The coupling.* The rotation $e^{-\mathrm{i}\theta\sigma_y/2} = \cos(\theta/2)\mathbb{1} - \mathrm{i}\sin(\theta/2)\sigma_y$ takes $|0\rangle$ to $\cos(\theta/2)\,|0\rangle + \sin(\theta/2)\,|1\rangle$. So for a system state $\psi = \psi_0\,|0\rangle + \psi_1\,|1\rangle$,

$$U_\theta(\psi \otimes |0\rangle) = K_0\psi \otimes |0\rangle + K_1\psi \otimes |1\rangle, \qquad K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \cos(\theta/2) \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sin(\theta/2) \end{pmatrix}.$$

2. *The probabilities.* By Born's rule for the projection $\mathbb{1} \otimes |k\rangle\langle k|$, the probe reads k with probability $\|K_k\psi\|^2 = \langle \psi, F_k\psi\rangle$, where $F_k = K_k^\dagger K_k$:

$$F_0 = |0\rangle\langle 0| + \cos^2(\theta/2)\,|1\rangle\langle 1|\,, \qquad F_1 = \sin^2(\theta/2)\,|1\rangle\langle 1|\,.$$

   The two matrices are positive and $F_0 + F_1 = \mathbb{1}$. For a density matrix ρ of the system, Theorem 3.4 on the pair gives $\mathrm{Tr}(\rho F_k)$.
3. *The state afterwards.* By Lüders' rule on the pair, a reading k of positive probability leaves the pair in the ray of $K_k\psi \otimes |k\rangle$, a product. The system is in the ray of $K_k\psi$.
4. *The two ends.* At $\theta = 0$, $F_0 = \mathbb{1}$. The probe always reads 0, and nothing is learnt. At $\theta = \pi$, $F_k = |k\rangle\langle k|$ and $K_k = |k\rangle\langle k|$, which is the ideal reading of σ_z.

For $0 < \theta < \pi$ the matrix F_1 has the eigenvalue $\sin^2(\theta/2)$, which is neither 0 nor 1, so F_1 is not a projection, and no observable has F_0 and F_1 as its eigenprojections. Exercise Sheet 11 shows that the state after the reading is not the one Lüders' rule gives either. The ideal model of Lecture 1 and Lecture 4 does not describe this measurement of the system, although the probe's reading is ideal. A family of positive matrices F_k that add up to $\mathbb{1}$, read with the probabilities $\mathrm{Tr}(\rho F_k)$, is called a *positive operator-valued measure*, or POVM. *Quantum Mechanics III: Foundations* studies such measurements and the states they leave.

The physics

In a hydrogen atom in its ground state, with no magnetic field, the electron's spin read alone is unpolarised, along every axis, although the two spins together are in the pure state ψ^-. Real detectors read a system through a probe, and many miss some of what they should detect. A photon counter of efficiency 0.9 detects each incident photon with probability 0.9, under its stated operating conditions. When it misses, its silence is the reading 0 of Example 11.11.

Part of a pair

The part: $\mathrm{Tr}(\rho_1 A) = \mathrm{Tr}\big(\rho\,(A \otimes \mathbb{1})\big)$; for a pure state $\rho_1 = CC^\dagger$, $\rho_2 = C^{\mathsf{T}}\overline{C}$ (Theorem 11.1). *Mixed parts:* ψ is entangled iff ρ_1 is mixed (Proposition 11.3). *Schmidt:* $\psi = \sum_{p_i>0} \sqrt{p_i}\, u_i \otimes v_i$ (Theorem 11.4). *No signalling:* $\mathbb{1} \otimes V$ and unrecorded readings of the second system leave ρ_1 (Proposition 11.8). *Purification:* $C = \sqrt{\rho}$ gives $\rho_1 = \rho$, $\rho_2 = \rho^{\mathsf{T}}$ (Theorem 11.10).

Summary.

- *The state of a part*: every experiment on one part is predicted by its reduced density matrix (Theorem 11.1).
- *No signalling*: nothing done to the second system, without a record, changes the first system's statistics (Proposition 11.8).
- *Purification*: every density matrix is the state of one part of a pure pair (Theorem 11.10).
- *How entangled*: the parts of a pure pair have one entropy, its measure of entanglement (Theorem 11.4).

The electron of ψ^- is unpolarised. Once the outcome of a reading of the proton's spin along an axis is known, the electron is in a pure state. Lecture 12 asks whether the readings of the two spins could have been fixed in advance.

References

Nielsen and Chuang, *Quantum Computation and Quantum Information*, §2.4.3, pp. 105–108, §2.5, pp. 109–111, §2.2.6, pp. 90–92, and §11.3, from p. 510 (with base-2 logarithms). Sakurai and Napolitano, *Modern Quantum Mechanics*, §3.4.5, pp. 176–180. *Linear Algebra for Physics*, Thm 10.7, Thm 10.13. d'Espagnat, *Conceptual Foundations of Quantum Mechanics* (1976).